

The Role of Active Touch in Designing Body Transformation Experiences

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Multisensory technologies have enabled a wide range of Body Transformation Experiences (BTEs) and demonstrated how visual, auditory, and haptic feedback can alter body perception and self-experience. However, many existing approaches rely on passive stimulation and controlled laboratory conditions, which limit their application in real-world settings. In the Body Transformation Experiences workshop, I aim to discuss the role of *active touch* in designing more interactive and robust BTEs. Active touch is defined as a self-generated and exploratory tactile interaction between the body and its surroundings that couples action and perception through continuous sensorimotor feedback. I argue that incorporating active touch into the design of multisensory systems can enhance user agency, support adaptation over time, and enable BTEs to emerge through engagement rather than imposition. Through examples grounded in haptic interaction, I illustrate opportunities to transform perceived body boundaries, effort, and control. I also highlight methodological implications, including the need for longitudinal studies and first-person approaches to capture lived experience.

Additional Key Words and Phrases: touch, haptics, active touch, body transformation

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1 Introduction

Body Transformation Experiences (BTEs) demonstrate that the perception of one's own body is not fixed but is dynamically shaped by multisensory processes. Research in Cognitive Neuroscience and Human–Computer Interaction (HCI) has shown that body representations, such as perceived size and shape, can be altered through carefully organized sensory feedback [7, 13, 14]. Classic paradigms such as the Rubber Hand Illusion demonstrate how synchronous visual and tactile stimulation can induce a compelling sense of ownership over an artificial limb [5]. Subsequent work has extended these findings to full-body illusions, changes in perceived body size, and transformations of self-location, highlighting the plasticity of body perception and its dependence on multisensory integration [4, 8].

Within HCI, these insights have inspired the design of multisensory technologies that intentionally elicit BTEs using visual, auditory, and haptic feedback [2, 6, 10]. Such systems have been explored in domains such as virtual reality, rehabilitation, and interactive art. However, a large portion of this work relies on passive stimulation paradigms, in which users receive pre-defined sensory inputs designed to induce specific perceptual outcomes. While effective in controlled settings, these approaches often limit user agency and may result in fragile or short-lived transformations that do not readily extend to everyday contexts.

In contrast, theories of sensorimotor contingency emphasize that perception arises through active engagement with the environment. From this perspective, the body is not a passive recipient of sensory input but an active participant in shaping perceptual experience through movement and interaction [1, 11, 12]. The concept of active touch further highlights how tactile perception is fundamentally tied to exploratory action, where sensing and actuation form a continuous closed-loop process [3, 9].

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Despite its relevance, active touch remains underexplored in the design of BTEs. In this position paper, I argue that active touch provides a paradigm for designing more robust BTEs. Building on this perspective, I discuss design opportunities for integrating active touch into multisensory systems, explain implications for evaluating BTEs over time, and highlight challenges related to agency and variability.

2 Active Touch in Body Transformation Experiences

In the context of BTEs, active touch provides a mechanism through which users may not only receive sensory feedback but also actively shape perceptions that give rise to altered body experiences. From this view, technologies that support BTEs should not merely deliver synchronized multisensory cues but should *enable users to discover and engage with new sensorimotor mappings*.

One design direction involves manipulating perceived body boundaries through active exploration. For instance, haptic feedback systems can dynamically alter perceived distances or resistances during reaching or probing movements. As users explore these altered mappings, they may experience shifts in perceived limb length (reachability) [7] and even a change in their limb movement speed [2]. Such interactions allow body transformations to emerge gradually through movement-based discoveries rather than immediate illusions.

Another opportunity lies in altering the perception of effort and physical capability. Haptic interfaces that modulate force feedback during manipulation tasks can change how heavy or resistant objects feel. Through repeated interaction, users may develop new expectations about the relation between their actions and resulting forces. This can potentially transform perceptions of strength, fatigue, or physical capability.

3 Implications for Evaluating Active BTEs

Designing BTEs containing active touch also raises new methodological considerations for evaluation. Conventional approaches often rely on short laboratory experiments combined with questionnaires assessing ownership, agency, or presence. While such methods are valuable, they may not fully capture the dynamic/adaptive nature of active touch interactions.

In this context, evaluations may include two major considerations. First, evaluation may analyze sensorimotor interaction patterns, such as exploration strategies, movement variability, or adaptation over time. These behavioral measures can provide insight into how users learn new sensorimotor mappings. Second, an evaluation may investigate how transformations stabilize or evolve with continued interaction. Because active touch involves learning and adaptation, some of the body transformations may emerge gradually as users develop familiarity with altered sensory contingencies.

4 Challenges: Agency and Variability

While active touch approaches offer promising directions for BTE design, they also introduce challenges that must be addressed. One important challenge concerns agency. Systems designed to transform body perception naturally shape how users experience their bodies. Designers must therefore ensure that users maintain a sense of control and intentionality during interaction. Maintaining transparency and allowing users to explore transformations voluntarily may help preserve the sense of agency.

A second challenge relates to variability across users and contexts. Because active touch relies on individual exploration strategies, users may interact with systems in highly diverse ways. As a result, BTE performance may vary

significantly between individuals, making it more difficult to achieve predictable or reproducible effects. Designing adaptable systems that respond to user behavior may help accommodate such variability.

Finally, active touch systems often require more complex sensing, actuation, and interaction design than passive stimulation systems. Therefore, the community's investment in developing robust hardware and software (such as sensors, processors, and actuators) can support these real-time and dynamic interactions in real-world environments.

5 Conclusion

Many existing BTEs rely primarily on passive sensory stimulation, limiting the role of user interaction in shaping real-world experiences. In this position paper, I argue that active touch provides a valuable framework for designing BTEs that emphasize exploration, engagement, and agency. By enabling users to actively interact with altered sensorimotor contingencies, multisensory technologies may support more robust and meaningful transformations of bodily experience. Integrating active touch into BTE design also opens new questions for evaluation and interaction design, particularly regarding adaptation over time and variability across users.

I hope this perspective contributes to ongoing discussions on how multisensory technologies can support richer and more sustainable bodily experiences, and encourages further exploration of active touch paradigms in the design of Body Transformation Experiences.

References

- [1] Ehud Ahissar and Eldad Assa. 2016. Perception as a closed-loop convergence process. *eLife* 5 (2016). doi:10.7554/eLife.12830
- [2] Easa AliAbbasi, Nihar Sabnis, Yuran Ding, Nadine Wagener, and Paul Strohmeier. 2025. Haptic Redirection: Modulating Hand Movement Speed with Vibrotactile Feedback. In *Proceedings of the Mensch Und Computer 2025* (Chemnitz, Germany) (*MuC '25*). Association for Computing Machinery, New York, NY, USA, 631–636. doi:10.1145/3743049.3748585
- [3] Easa AliAbbasi, Gabriela Vega, Dennis Wittchen, and Paul Strohmeier. 2025. Physical Compliance and the Compliance Illusion: The Importance of Action for Perception. In *2025 IEEE World Haptics Conference (WHC)* (Suwon, Korea). 349–356. doi:10.1109/WHC64065.2025.11123193
- [4] Olaf Blanke. 2012. Multisensory brain mechanisms of bodily self-consciousness. *Nature Reviews Neuroscience* 13, 8 (2012), 556–571. doi:10.1038/nrn3292
- [5] Matthew Botvinick and Jonathan Cohen. 1998. Rubber hands ‘feel’ touch that eyes see. *Nature* 391, 6669 (1998), 756–756. doi:10.1038/35784
- [6] Amar D'Adamo, Marte Roel Lesur, Laia Turmo Vidal, Mohammad Mahdi Dehshibi, Daniel De La Prida, Joaquín R. Díaz-Durán, Luis Antonio Azpicueta-Ruiz, Aleksander Väljamäe, and Ana Tajadura-Jiménez. 2024. SoniWeight Shoes: Investigating Effects and Personalization of a Wearable Sound Device for Altering Body Perception and Behavior. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). Association for Computing Machinery, Article 93, 20 pages. doi:10.1145/3613904.3642651
- [7] Yuran Ding, Nihar Sabnis, and Paul Strohmeier. 2024. Motionless Movement: Towards Vibrotactile Kinesthetic Displays. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '24*). ACM, New York, NY, USA, Article 418. doi:10.1145/3613904.3642499
- [8] Konstantina Kiltani, Raphaëla Groten, and Mel Slater. 2012. The sense of embodiment in virtual reality. *Presence: Teleoperators and Virtual Environments* 21, 4 (2012), 373–387. doi:10.1162/PRES_a_00124
- [9] Roberta L. Klatzky. 2025. Haptic Perception and Its Relation to Action. *Annual Review of Psychology* 76, Volume 76, 2025 (2025), 227–250. doi:10.1146/annurev-psych-011624-101129
- [10] Zhipeng Li, Yishu Ji, Ruijia Chen, Tianqi Liu, Yuntao Wang, Yuanchun Shi, and Yukang Yan. 2025. Modeling the Impact of Visual Stimuli on Redirection Noticeability with Gaze Behavior in Virtual Reality. In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (*CHI '25*). Association for Computing Machinery, New York, NY, USA, Article 38, 18 pages. doi:10.1145/3706598.3713392
- [11] Alva Noë. 2004. *Action in perception*. MIT press.
- [12] J. Kevin O'Regan and Alva Noë. 2001. A sensorimotor account of vision and visual consciousness. *Behavioral and Brain Sciences* 24, 5 (2001), 939–973. doi:10.1017/s0140525x01000115
- [13] Ana Tajadura-Jiménez, Torsten Marquardt, David Swapp, Norimichi Kitagawa, and Nadia Bianchi-Berthouze. 2016. Action sounds modulate arm reaching movements. *Frontiers in psychology* 7 (2016), 1391. doi:10.3389/fpsyg.2016.01391
- [14] Laia Turmo Vidal, Ana Tajadura-Jiménez, José Manuel Vega-Cebrián, Judith Ley-Flores, Joaquín R. Díaz-Durán, and Elena Márquez Segura. 2024. Body Transformation: An Experiential Quality of Sensory Feedback Wearables for Altering Body Perception. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork, Ireland) (*TEI '24*). ACM, New York, NY, USA, Article 25. doi:10.1145/3623509.3633373

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